MIL-I-83117B(USAF)
29 August 1975
SUPERSEDING
MIL-I-83117A(USAF)
12 July 1973

MILITARY SPECIFICATION

INDICATORS, ANGLE OF ATTACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for 26-V, 400 Hz local angle of attack indicators.

1.2 Classification. Indicators shall be of the following types as specified (see 6.2):

Type	Range
ARU-45/A	0 to 30 units
ARU-46/A	10 to 50 degrees

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges

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MIL-T-25627	Transmitter, Synchro, Angle of Attack or Side Slip
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic
MIL-C-83723	Connector, Electric, Circular, Environment Resisting, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard, General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-889	Dissimilar Metals
MS27572	Case, Instrument, Flange-Mounted, Aircraft
MS27582	Indexer, Angle of Attack
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions may be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

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3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used they shall be treated with a fungicidal agent acceptable to the procuring activity.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts, except where magnetic materials are essential.

3.3.3 Nonferrous materials. Nonferrous materials shall be used for all parts, except where ferrous materials are essential.

3.3.4 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.4.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Deleterious fumes. Materials as installed in the indicator and under the service conditions specified herein shall not liberate deleterious fumes.

3.3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction

3.4.1 Design. The indicator dimensions shall be as shown on figures 1 and 2. The indicator shall be designed for use with a transmitter conforming to MIL-T-25627 to provide an indication of the aircraft angle of attack. The indicator shall incorporate electronic switches for energizing a set of indicator lamps conforming to MS27582. The set of lamps shall consist of three separate 28-V dc lamps to indicate to the pilot the low, medium-low, medium, medium-high, and high ranges of angle of attack. Separately, the three lamps shall indicate low, medium, and high ranges of angle of attack. The medium-low indication shall be obtained by simultaneously energizing the lamps that

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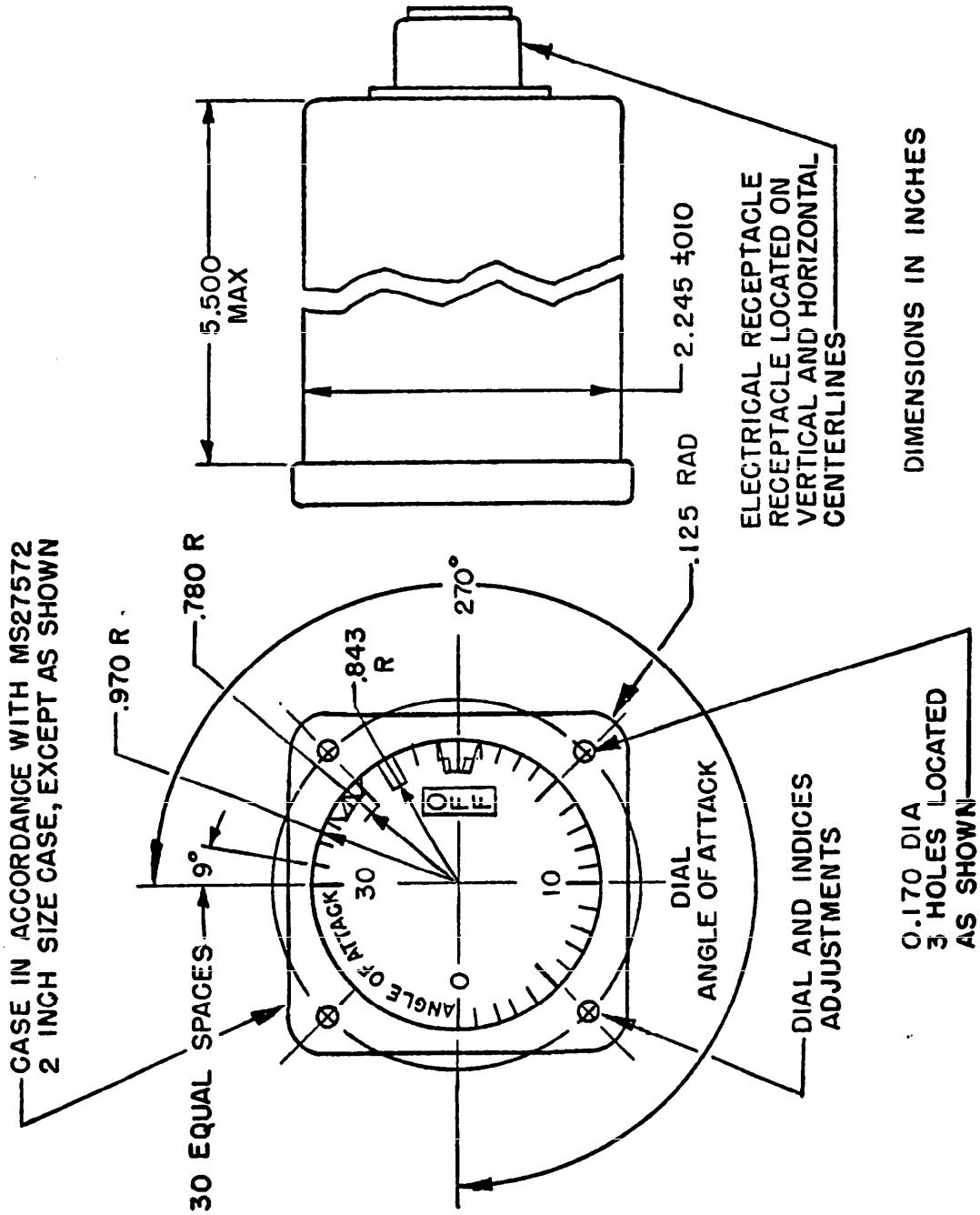


FIGURE 1. ARU-45/A Indicator

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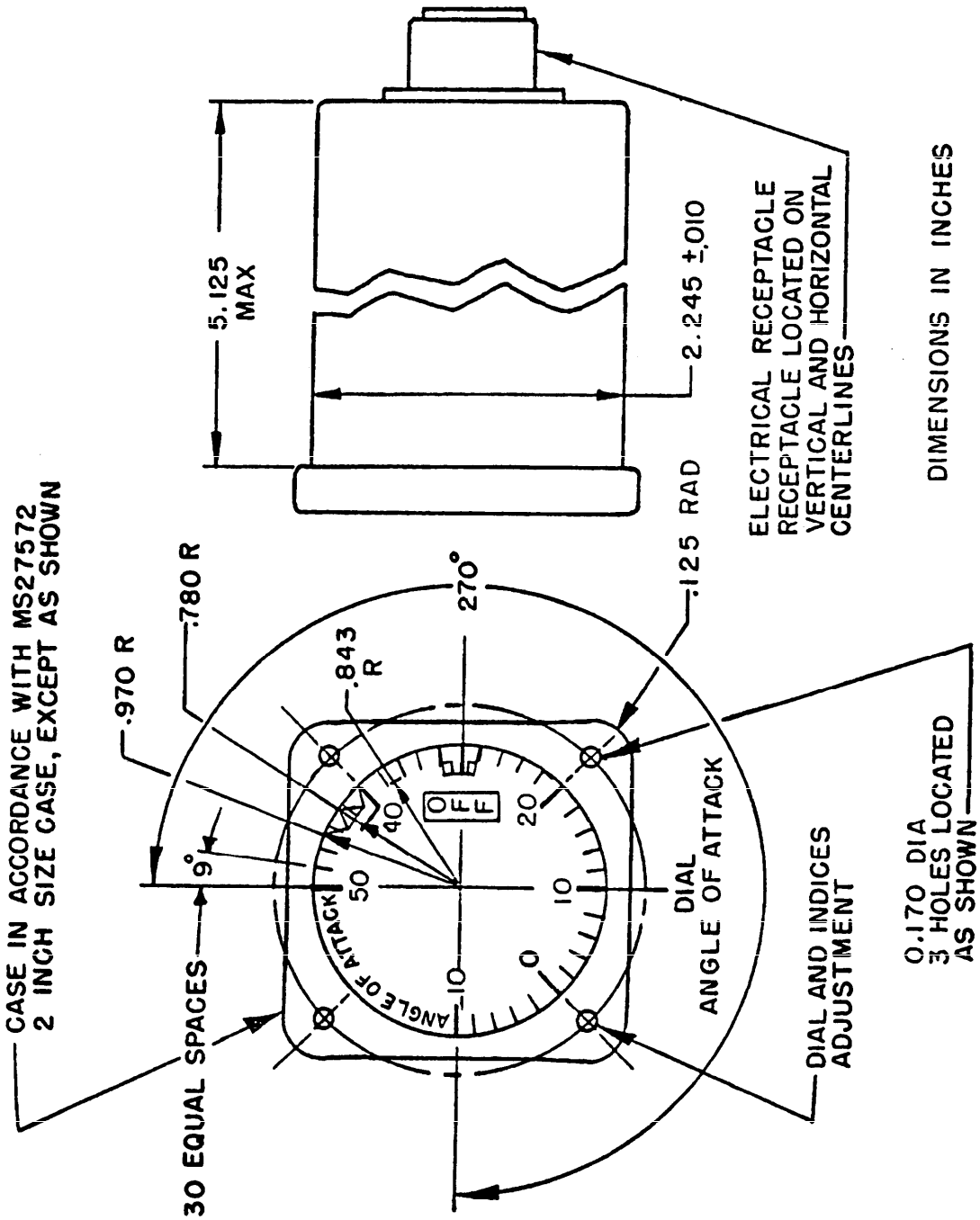


FIGURE 2. ARU-46/A Indicator

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separately indicate the low range and the medium range. The medium-high indication shall be obtained by simultaneously energizing the lamps that separately indicate the high range and the medium range. On preproduction indicators, the indexer light switching interval shall be variable by means of adjustments on the back of the case. The procuring activity shall use preproduction indicators to select a specific value for each aircraft. The specific value shall be set as a constant on production indicators and the external adjustments shall be eliminated. An electronic switch shall also be provided for stall warning output.

3.4.1.1 Reliability. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,500 hours at a 90 percent confidence level.

3.4.1.2 Longevity. The indicator shall have an operating life (equipment longevity as defined in MIL-STD-781) of 3,000 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF.

3.4.1.3 Maintainability. Maintainability shall be in accordance with MIL-STD-471.

3.4.1.3.1 Qualitative requirements. The following qualitative requirements shall be considered in the design of the indicator:

a. Minimization of complexity of maintenance tasks (for example: Calibration, adjustments, inspection, et cetera) by maximum use of simple design which includes optimum interchangeability and use of the standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective part, assembly, or component

d. Minimum number and types of tools and test equipment (special and standard) needed to perform maintenance

e. Optimum accessibility in all components requiring maintenance, inspections, removal, or replacement.

3.4.1.3.2 Corrective maintenance downtime. The mean corrective maintenance downtime (M_{ct}) shall be 100 minutes less the time required to unseal and then hermetically seal the instrument after repair. The maximum corrective maintenance downtime ($M_{max_{ct}}$) shall be no greater than 300 minutes less the time required to unseal and seal the case.

3.4.1.4 Safety. The indicator shall present no danger, injury or hazard to operating and service personnel.

3.4.1.4.1 Hazard analysis. A safety hazard analysis shall be performed to determine, from a safety consideration, the functional relationship of components and equipments interfacing with the indicator. The safety analysis shall identify all components, equipments, and materials whose performance degradation or functional failure could result in category III and IV hazards as defined in MIL-STD-882, paragraph entitled Hazard Level. This analysis shall include a determination of the failure modes and the effects on safety

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should failures occur. Areas to be considered are the appropriate sections of MIL-STD-882, as listed under paragraph entitled System Hazard Analyses, and the effects of storage, shelf life, transportation, and packaging. Where practicable, design changes shall be made to eliminate or minimize the hazards. If the hazards cannot be eliminated, alternative controls such as recommended changes to interfacing equipment shall be presented to the procuring activity for resolution.

3.4.2 Construction. The indicator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.2.1 Overhaul. The indicator shall be so constructed that overhaul, including replacement of worn parts, adjustment balance, and calibration, may be accomplished by instrument and electronic repairmen provided with simplified tools and equipment and trained in the repair of electronic equipment.

3.5 Performance. The indicator shall meet all requirements specified herein after exposure to the following conditions:

a. Operating temperatures ranging from -54° to +71°C (-65° to +160°F)

b. Altitudes from sea level to 70,000 feet

c. Relative humidity up to 100 percent

d. Rainfall of 4 ±1 inches each hour

e. Fungus growth as encountered in tropical climates

f. Exposure to salt-laden atmosphere

g. Sand and dust particles as encountered in desert areas

h. Vibration per 4.6.13

i. Acceleration force of 2g for 1 minute in the forward direction

j. Application of 250V dc for 10 seconds.

3.5.1 Dynamic response. The indicator shall be designed to have a damping ratio between 0.8 and 1.8 and a slew rate of 8 units (degrees)/seconds at 400 Hz. The frequency response to sinusoidal inputs shall be such that the indicator output shall be attenuated no more than 6 db at 0.2 Hz and no less than 24 db at 2.0 Hz. On preproduction indicators, the damping shall be variable within the limits specified by means of adjustments on the back of the case. The procuring activity shall use the preproduction indicators to select a specific value for each aircraft. The specific value selected shall be set as a constant on production indicators, and the external adjustments shall be eliminated.

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3.5.2 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 as specified for class A1 equipment.

3.5.3 Attitude. The indicator shall be capable of operating satisfactorily in any attitude.

3.5.4 Scale and friction error. The indicator shall operate within the scale and friction error tolerances specified in table I and II.

TABLE I. Transmitter Settings and Scale Error Tolerances

Indicator Reading		Scale Error Tolerances	
ARU-45/A (units)	ARU-46/A (degrees)	Room High and Low Temperature (units)	(degrees)
0	-10	.25	.5
5	0	.25	.5
10	10	.25	.5
15	20	.25	.5
20	30	.25	.5
25	40	.25	.5
30	50	.25	.5

TABLE II. Friction Error Tolerances

Test	Tolerance	
	Units	Degrees
Friction error		
Room temperature	0.25	0.50
High temperature	0.25	0.50
Low temperature	0.25	0.50
Position error	0.25	0.50

3.5.4.1 Sensitivity. The indicator shall provide a noticeable change in reading with an input change of 0.25 degree in the transmitter input.

3.5.5 Magnetic effect. When the indicator is held in various positions magnetically east or west of the not more than 12 inches from the center of a free magnet in a magnetic field having a horizontal intensity of 0.18 ±0.01 oersted, deflection of the magnet shall not exceed 5°.

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3.6 Electronic requirements. Electronic requirements shall be in accordance with MIL-E-5400, except as specified herein.

3.7 Modern techniques. Modern techniques of miniaturization, potting, and hermetic sealing shall be used to the greatest extent practicable without sacrificing ruggedness, reliability, or service life.

3.8 Case. The case shall be as shown on figures 1 and 2.

3.8.1 Body and mounting flange. The body and mounting flange of the case shall be metal, uniform in texture, shall have a smooth surface, and shall be finished in durable lusterless black conforming to FED-STD-595, color No. 37038. The inside of the case around the electrical connector shall be of sufficient strength to prevent being damaged when the electrical connection is tightened during installation.

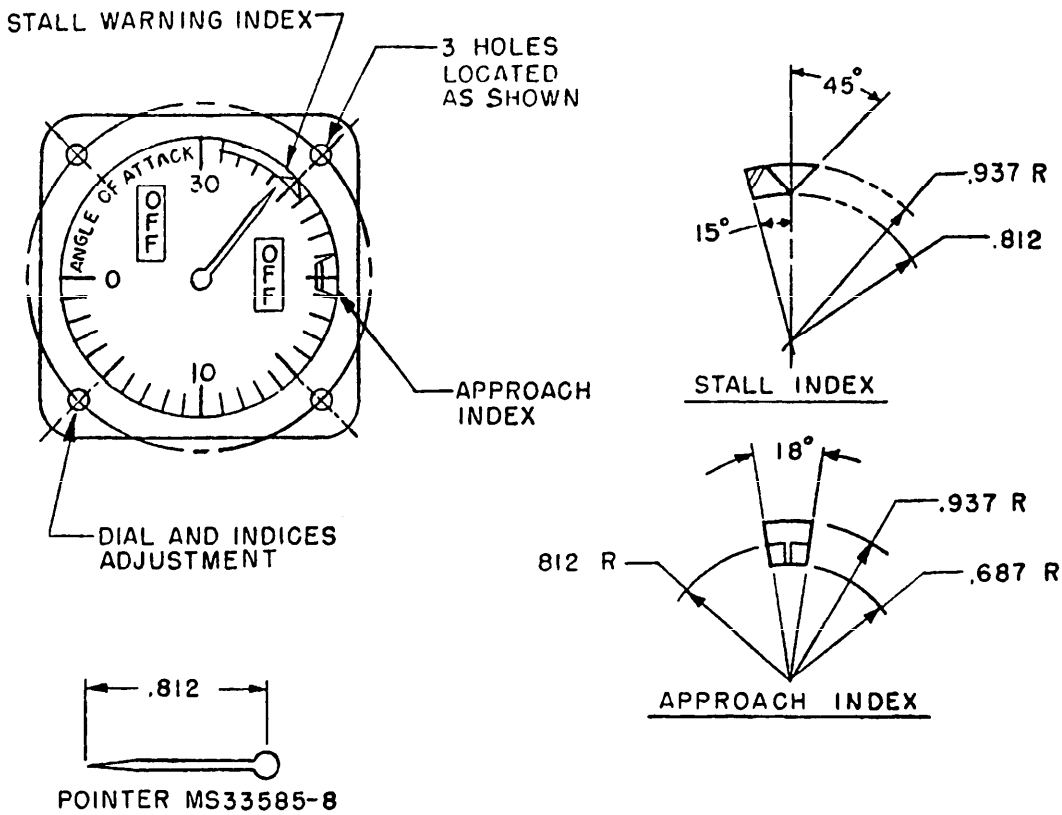
3.8.2 Indices. The indicator shall be designed to include as a minimum a stall warning index and an approach index. Relative positions of the indices shall be as shown on figures 3 and 4. Provisions shall be included to permit external adjustment of the dial within ±10° angle of attack (±10 units angle of attack) from the position shown on figures 3 and 4. Adequate stops shall also be provided to prevent the pointer from making a full rotation. The approach index shall be fixed at the 3 o'clock position on the indicator. The dial shall be adjusted relative to the fixed approach index. The stall warning index shall be internally adjustable but shall remain fixed to the dial during external adjustment of the dial. The stall warning index shall be adjustable within ±20 degrees angle of attack (10 units angle of attack) from the vertical (12 o'clock) position. The indicator shall be so designed that manual adjustment of the dial will cause all switching points to remain fixed relative to the 3 o'clock position on the indicator.

3.8.3 Switches. The indicator will provide output switches for indexer light operation and stall warning signal. The switch points shall be repeatable to within ±0.25 units (±0.50 degrees). The switch point hysteresis shall not exceed 0.1 ±0.05 units (0.2 ±0.1 degrees).

3.9 Dial. The dial shall conform to figures 1, 2, 3, and 4. Dial markings shall be in accordance with figures 3 and 4.

3.9.1 Cover glass. The cover glass shall be provided with a coating conforming to MIL-C-14806.

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NUMERALS SHALL DISTINCTLY INDICATE THE GRADUATION TO WHICH EACH APPLIES

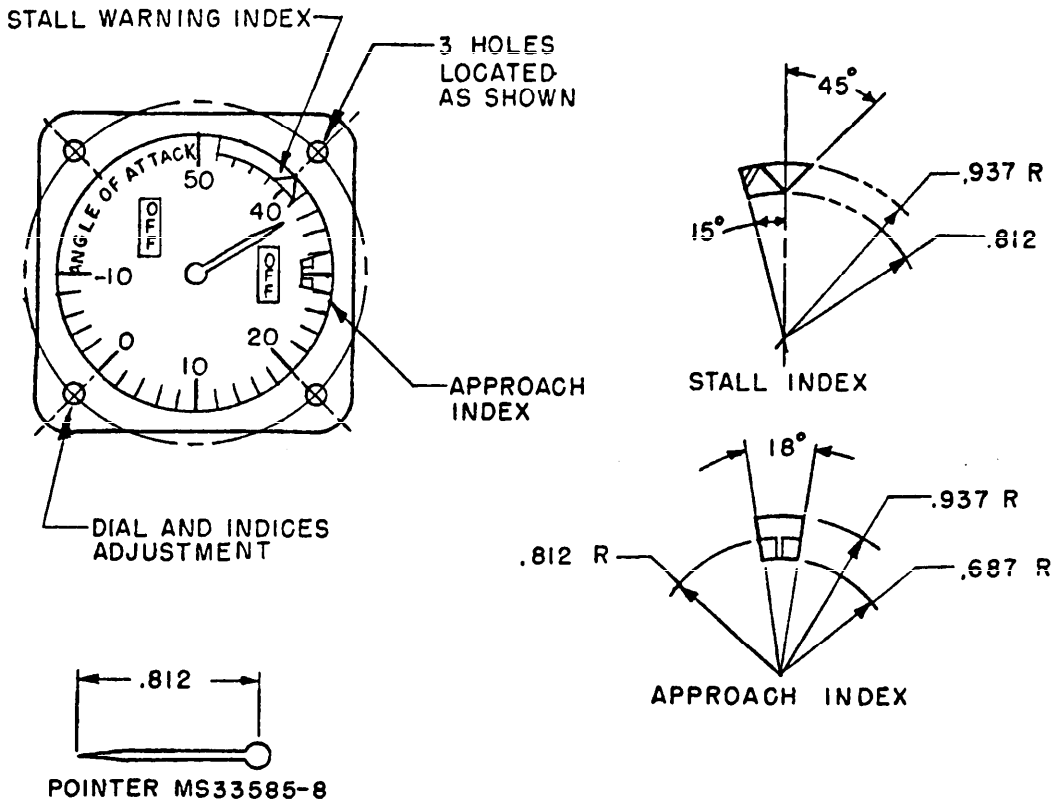
DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED, TOLERANCES: ±.010, ANGLES ±1°

INDICES REFERENCE

MARKING	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION ±.005	MATERIAL OR FINISH
MAJOR GRADUATIONS	---	.040	LUSTERLESS WHITE COLOR NO. 37875 OF FED-STD-595
INTERMEDIATE GRADUATIONS	---	.023	
NUMERALS	.187	.023	
LETTERS	.125	.020	
SHADED PORTION OF POINTER AND REFERENCE INDICES	---	---	
BACKGROUND OF DIAL UNSHADED PORTION OF POINTER AND REFERENCE INDICES	---	---	LUSTERLESS BLACK COLOR NO. 37038 OF FED-STD-595

FIGURE 3. Dial and Pointer Details (ARU-45/A)

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NUMERALS SHALL DISTINCTLY INDICATE THE GRADUATION TO WHICH EACH APPLIES

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ±.010, ANGLES ±1°

MARKING	INDICES REFERENCE: HEIGHT OR LENGTH ±.010	INDICES REFERENCE: WIDTH OF LINE OR GRADUATION ±.005	MATERIAL OR FINISH
MAJOR GRADUATIONS	---	.040	LUSTERLESS WHITE COLOR NO. 37875 OF FED-STD-595
INTERMEDIATE GRADUATIONS	---	.023	
NUMERALS	.187	.023	
LETTERS	.125	.020	
SHADED PORTION OF POINTER AND REFERENCE INDICES	---	---	
BACKGROUND OF DIAL UNSHADED PORTION OF POINTER AND REFERENCE INDICES	---	---	LUSTERLESS BLACK COLOR NO. 37038 OF FED-STD-595

FIGURE 4. Dial and Pointer Details (ARU-46/A)

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3.9.2 Viewing angle. The pointer, numerals, and at least 1/32 inch of the shortest graduations, and all indices on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 20 degrees with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.10 Pointer. The pointer shall be in accordance with figures 3 and 4.

3.11 Standard synchro test transmitter. A standard calibrated synchro, Clifton Precision Products type CGC-10-AS-7 or electrical equivalent, with minimum and known errors shall be provided. A 0° to 360° dial, clamped to the rotor or the synchro, shall be settable, rotatable, and readable through 360° of rotation.

3.11.1 Indicator zeroing and sensing. A synchro such as that used in the angle of attack transmitter conforming to MIL-T-25627 shall be connected to the indicator in accordance with table III. Pin connections specified in table III shall apply when the transmitter is installed on the right hand side of the aircraft. The procedure shall then be as follows:

a. Label the rotor leads H and C.

b. Select the Z stator lead and connect it to the H rotor lead.

c. Apply 26V 400-Hz power to rotor leads H and C. Connect a VTVM between the remaining stator leads (x and y), and rotate the rotor or stator until a NULL or minimum voltage is obtained.

d. Measure the voltage between the H lead and the remaining stator leads. If this voltage is less than the excitation voltage of the rotor, the synchro is at LOW NULL. If the voltage is greater than the rotor excitation voltage, the synchro is at HIGH NULL.

e. Turn the synchro to a low null position. Connect a VTVM across one of the remaining stator leads and pin A (lead C) with the second lead open. When the vane is turned to indicate increasing angle of attack, the voltage will increase before decreasing if the X stator lead is connected. The voltage will decrease before increasing if the Y stator lead is connected. All leads of the synchro shall be labeled in accordance with the results of the above tests.

f. The indicator shall read 15 ±0.2 units (30 ±0.2 degrees) when the transmitter synchro is at low null and shall increase in reading when the transmitter is rotated toward increasing angle of attack.

3.12 Power. The indicator shall operate on 26V 400-Hz single-phase power supplied from an aircraft infinite bus source having characteristics conforming to MIL-STD-704.

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TABLE III. Indicator Connections

Connector Pins	Functions
A B	Neutral (external power ground) 26V 400 Hz
C(X) D(Y) E(Z)	Indicator servo control transformer stator leads
F G H J K L	High light (V) Center light (O) Low light (A) Common 28V power input Spare } Indexer Lights
M N	White lighting (positive 5V dc) Ground
P R	Stall warning (28V 3 amp resistive)
S T U V	Spares

3.12.1 Power consumption. Power consumption shall not exceed 25 va.

3.12.2 Power variation. The indicator shall operate properly and continuously at voltages ranging from 23 to 28V and frequencies ranging from 320 to 480 Hz, or any combination thereof, and under the transient power conditions specified in MIL-STD-704.

3.12.3 Power-off flag. A power-off flag as shown on figures 1, 2, 3, and 4 shall be incorporated. The flag shall be black lettering on a red background and shall be so mechanized that it will appear when power to the indicator is interrupted.

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3.13 Connectors. Electrical connectors shall be in accordance with MIL-C-83723 and shall be located as shown on figures 1 and 2.

3.14 Integral lighting. The indicator shall be integrally lighted in accordance with MIL-L-27160.

3.14.1 Prime standard. Integral lighting shall be comparable to and shall balance in brightness with a prime standard which shall be an indicator conforming to this specification or a lighted model. The specific voltage which the procuring activity has established as necessary to assure equality of presentation (from a lighting standpoint) in adjacent but dissimilar instruments shall be applied to the prime standard. The contractor shall submit to the procuring activity the procedures for determining compliance with this requirement.

3.15 Jewel bearings. When synthetic jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.16 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of *MIL-STD-100* shall govern the manufacturer's part numbers and changes thereto.

3.17 Weight. The weight of the indicator shall not exceed 2 pounds.

3.18 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.19 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454.

3.20 Lubrication. The indicator shall be lubricated in accordance with MIL-STD-838.

3.21 Finishes and protective coatings

3.21.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.21.1.1 Dials, small holes, and case inserts need not be anodized.

3.21.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.21.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

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3.21.1.4 Castings containing nonaluminum-alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.21.1.5 When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.21.2 Cadmium plating. Cadmium plating shall be in accordance with type I, class 3 of QQ-P-416.

3.22 Installation

3.22.1 Mounting screws. The contractor shall furnish sufficient mounting screws for installing the indicator. They shall be round-head, brass machine screws and shall have a durable dull-black finish. Length shall be sufficient to permit mounting on a panel 3/16-inch thick. Screw threads shall be size 6-32 UNF-2A.

3.22.2 Envelope. An envelope containing the mounting screws shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS

3.23 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.24 Workmanship. The indicator, including all parts and accessories, shall be so fabricated and finished as to assure freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts.

3.24.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.24.2 Cleaning. The indicator shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign materials removed during and after final assembly.

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3.24.3 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.24.4 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Preproduction tests . . . . . See 4.4

b. Quality conformance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature, approximately 25°C (77°F).

4.3.2 Voltage input. Unless otherwise specified, the indicator shall be tested with 26V, 400-Hz (nominal) single-phase power connected to the specific pins of the indicator connector.

4.3.3 Proper connection. When the indicator is referred to as being properly connected, it shall be connected to a test fixture that provides all power and signals as shown in table III.

4.3.3.1 The test specified in 4.6.3 may be performed, at the contractor's option, prior to final sealing of the indicator.

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4.3.4 Visual observation. Unless otherwise specified, all visual observations of indicator performance shall be made at a distance of 2 feet.

4.3.5 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of nine indicators representative of the production equipment, except for the damping identified with the manufacturer's own part number and such other information as required by the procuring activity. Three of the indicators shall be subjected to environmental tests while the remaining six indicators shall be subjected to reliability tests except for the tests specified in 4.6.19.4.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831.

4.4.3 Preproduction tests. Preproduction tests shall consist of all the tests specified in 4.6.

4.5 Quality conformance tests. Quality conformance tests shall consist of the following:

a. Individual tests . . . . . See 4.5.1

b. Sampling tests . . . . . See 4.5.2.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Synchro and receiver zero . . . . . See 4.6.2

c. Servo operation tests . . . . . See 4.6.3

d. Scale error and friction test . . . . . See 4.6.4

e. Power-off flag test . . . . . See 4.6.5

f. Early failure detection test . . . . . See 4.6.7

g. Lighting tests . . . . . See 4.6.6

h. Slew rate test . . . . . See 4.6.3.3

i. Switch point test . . . . . See 4.6.4.2

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4.5.2 Sampling tests

4.5.2.1 Sampling plan A. One indicator shall be selected at random from the first ten on the contract or order, two indicators from the next ninety, and two from each additional one hundred produced. The indicators so selected shall be subjected to the following tests as described herein and under 4.6:

a. Individual tests . . . . . See 4.5.1

b. Power consumption test . . . . . See 4.6.8

c. Voltage variation tests . . . . . See 4.6.9

d. Magnetic effect test . . . . . See 4.6.10

e. Acceleration test . . . . . See 4.6.11

f. Dielectric test . . . . . See 4.6.12

g. Vibration error test . . . . . See 4.6.13

h. Servo null position shift test . . . . . See 4.6.14

i. Low-temperature operation test . . . . . See 4.6.15

j. High-temperature operation test . . . . . See 4.6.16

k. Sampling plan A tests of MIL-L-27160

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three indicators shall be selected at random from the first fifteen on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests . . . . . See 4.5.2.1

b. High temperature exposure test . . . . . See 4.6.18.1

c. Low temperature exposure test . . . . . See 4.6.18.2

d. Temperature-altitude (cycling) test . . . . . See 4.6.18.3

e. Temperature shock test . . . . . See 4.6.18.4

f. Humidity test . . . . . See 4.6.18.5

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g. Fungus test . . . . . See 4.6.18.6

h. Salt-fog test . . . . . See 4.6.18.7

i. Dust test . . . . . See 4.6.18.8

j. Vibration failure test . . . . . See 4.6.18.9

k. Rain test . . . . . See 4.6.18.10

l. Acceptance reliability . . . . . See 4.6.19.4

m. Sampling plan B tests of MIL-L-27160

4.5.2.2.1 At the option of the contractor, the tests specified in 4.5.2.2 may be divided into three groups as shown in table IV to be performed on three sets of three indicators if a need exists to expedite tests. Each indicator shall meet the individual tests after completing the group of specific tests to which it is subjected.

TABLE IV. Groups and Tests

Group	Test	Paragraph
I	High-temperature exposure	4.6.18.1
	Low-temperature exposure	4.6.15
	Temperature shock	4.6.18.4
II	Humidity	4.6.18.5
	Fungus	4.6.18.6
	Salt-fog	4.6.18.7
	Dust	4.6.18.8
	Rain	4.6.18.10
III	Vibration failure	4.6.18.9
	Acceleration	4.6.11
	Acceptance reliability	4.6.20.4

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4.5.2.3 Rejection and retest. When one indicator selected from a production run fails to meet the specification, indicators still on hand or later produced shall not be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that the indicators meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in indicators already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. Prior to sealing, the indicator shall be inspected to determine compliance with the requirements of 3.3, 3.4.2, 3.18, 3.19, 3.20 and 3.24. If these inspections are performed as part of the assembly inspections, they shall be documented during the equipment assembly and this documentation shall be checked as part of the Examination of Product. After sealing, filling and painting, the indicator shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, markings, weight, dimensions and visibility of the display components. The case shall be inspected for proper painting and general good workmanship and the connector shall be inspected to make sure that no pins are bent or damaged.

4.6.2 Synchro zero test. The CT receiver synchro shall be checked for zero and direction in accordance with the method specified in section 3.

4.6.3 Servo operation tests

4.6.3.1 Hunting and jumping tests. There shall be no noticeable hunting or jumping when the standard calibrated synchro shaft is not rotated. Under dynamic conditions, the indicator motion shall be smooth and continuous without hunting and jumping.

4.6.3.2 Servo sensitivity and friction error tests. The indicator shall be properly connected. The rotor of the calibrated synchro shall be rotated 0.6° and there shall be a perceptible movement of the pointer as the CT transmitter is excited.

4.6.3.3 Slew rate test. The indicator shall be subjected to a 0 to 30 unit (-5 to 25 degree) step function input. The time required for the pointer to move from 5 to 25 units (0 to 40 degrees) shall be 2.5 ±0.5 seconds.

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4.6.4 Scale error and friction test. The indicator shall be properly connected to a standard transmitter and the shaft of the transmitter rotated through the test settings specified in table I. The indicator scale error shall not exceed the tolerances specified therein. At each setting, two indicator readings shall be taken, the first before the indicator has been tapped and the second after the indicator has been tapped. The difference between the two readings shall constitute friction which shall not exceed the tolerances specified in table II.

4.6.4.1 Position error test. The reading of the indicator pointer taken when the indicator is in the normal upright position shall not differ from the reading when the indicator is in any other position by more than the values specified in table II.

4.6.4.2 Switch point test. All switches shall be actuated by slowly varying the input synchro signal. The switch point accuracy shall be ±.25 units (±0.5 degrees). The difference between increasing angle of attack switch points and decreasing angle of attack switch points shall be 0.1 ±0.05 units (0.2 ±0.1 degrees).

4.6.5 Power-off flag test. With the indicator properly connected but no AC power applied, the power-off flag shall be in full view. AC power shall be applied, and the flag shall immediately disappear.

4.6.6 Lighting tests. Lighting tests shall be conducted in accordance with the individual lighting tests of MIL-L-27160.

4.6.7 Early failure detection test. Following final assembly and after completion of all other individual tests, the indicator shall be properly connected to a standard transmitter with power applied to the servo system and the lighting circuit. The angle of attack transmitter shall be rotated through the complete indicator range at a rate of five cycles per hour. This test shall be conducted for a minimum of 5 hours. No failures shall occur.

4.6.8 Power consumption test. The indicator power consumption shall be measured with suitable meters and shall not exceed 25 va.

4.6.9 Voltage variation tests. The indicator shall be properly connected with power applied and subjected to the transient power conditions specified in MIL-STD-704. The indicator shall not be damaged. The indicator shall then be subjected to the individual tests at voltages and frequencies of 23V 320 Hz and 28V 480 Hz, except the scale error accuracy may be increased from .5 units and servo slew speed may vary by 20 percent from the nominal 2.5 seconds in the 320- to 480-Hz range.

4.6.10 Magnetic effect test. This test shall be made first with no power applied to the indicator and then with the indicator operating on rated power. The indicator shall be held in various positions magnetically east or west of and 12 inches from the center of a free magnet approximately 1-1/2 inches long in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted. The maximum deflection of the magnet shall not exceed 5°. An aircraft compass with the compensating magnets removed may be used as the free magnet for this test.

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4.6.11 Acceleration tests. The acceleration test shall be conducted in accordance with MIL-STD-810, Method 513.2, procedures I and II for aircraft equipment, except where noted herein.

4.6.11.1 Structural test. The test time duration in each direction shall be at least one minute following centrifuge stabilization. After completion of the test the unit shall be subjected to and shall meet the individual test.

4.6.11.2 Operational test. The unit shall be operated with power applied. The test shall be performed with the indicator set at 0, 10, 20, and 30 units (i.e. - 10, 10, 30, and 50 degrees). The change in reading during the test shall not exceed ±0.25 units (±0.50 degrees).

4.6.12 Dielectric test. A potential of 250V dc shall be applied between isolated pins and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation.

4.6.13 Vibration error test. The indicator shall be properly connected with power applied and tested in accordance with MIL-STD-810, Method 514.2, procedure IA, except the power spectral density (G^2/Hz) shall be as shown in figure 5. The indicator shall be considered hard mounted in service. During this test the indicator shall be set at 15 units (30 degrees) and the reading shall not vary by more than ±0.25 units (0.50 degrees) during vibration.

4.6.14 Servo null position shift test. The servo shall be properly connected to the standard test transmitter which shall be set at the zero indication. The input a-c voltage and frequency shall be varied within the limits of 23 to 28V and 320 to 480 Hz, or any combinations thereof. Zero indication of the servoed pointer shall not shift more than 0.1 units (0.2 degrees) from the normal null position.

4.6.15 Low-temperature operation test. The indicator shall be properly connected with power applied and subjected to a temperature of -54° ±2°C (-65° ±4°F) for a period of 4 hours. At the end of this period and while still at the specified temperature, power shall be applied. There shall be no sticking of movable component parts, and the pointer shall move throughout its range of travel without sticking. No failure shall result from this test. There shall be a 5-minute warmup period after power is applied before data are taken.

4.6.16 High-temperature operation test. With power applied, the indicator shall be properly connected, placed in a chamber, and maintained at a temperature of 71° ±2°C (160° ±4°F) for a period of 4 hours. There shall be no sticking of movable component parts, the pointer shall move throughout its range of travel without sticking, and there shall be no failure as a result of this test. The indicator shall then pass the tests specified in 4.6.3.1 through 4.6.6 except that the tolerances shall be as given in table I.

4.6.17 Electromagnetic interference test. The electromagnetic interference test shall be conducted in accordance with MIL-STD-462. Test methods shall be those specified in MIL-STD-461 for class A1 equipment.

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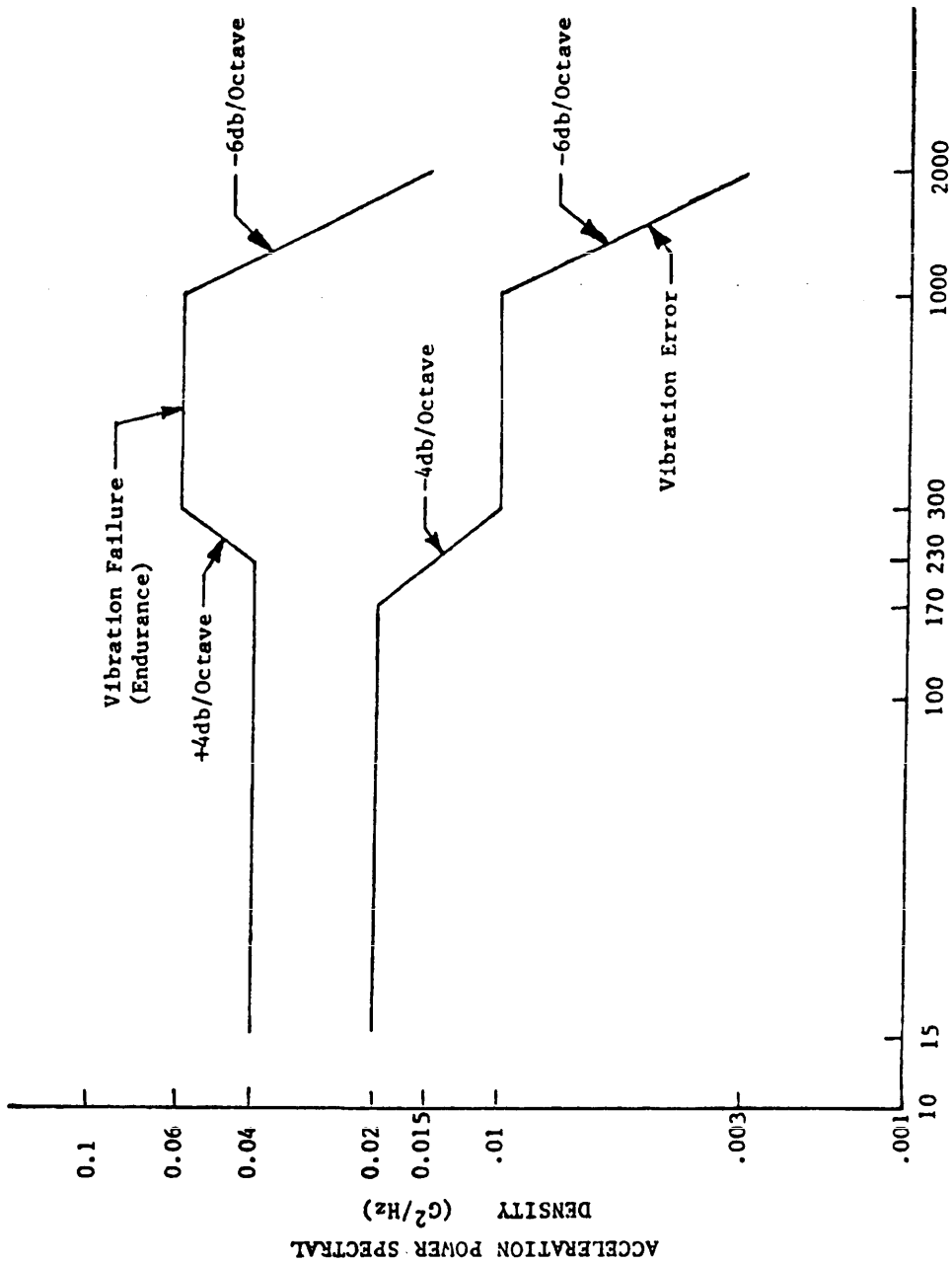


FIGURE 5. Random Vibration Envelope

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4.6.18 Environmental tests. The indicator shall be subjected to the following environmental tests, performed in accordance with the applicable procedures of MIL-STD-810 and as specified herein.

4.6.18.1 High temperature exposure test. The indicator shall be properly connected and subjected to a high temperature exposure test in accordance with method 501.1, procedure I. No power shall be applied. At the end of the 48-hour exposure period and while still at the specified temperature, the indicator shall pass the tests specified in 4.6.3.1 through 4.6.6 except that the tolerances shall be as specified in table I.

4.6.18.2 Low temperature exposure test. The indicator shall be properly connected and subjected to a low temperature exposure test in accordance with method 502.1, procedure I with no power applied. At the end of the 48-hour exposure period and while still at the low temperature, the indicator shall pass the tests specified in 4.6.3.1 through 4.6.6 except that the tolerances shall be as specified in table I.

4.6.18.3 Temperature-altitude (cycling) test. The indicator shall be subjected to a temperature-altitude cycling test in accordance with method 504.1, procedure I, except that step 7 shall be conducted at 71°C (160°F) and step 10 shall not be conducted. There shall be no binding or sticking at any time. After the indicator has returned to room temperature, it shall pass the individual tests. After the application of power and before data are taken, there shall be a 5-minute warmup period.

4.6.18.4 Temperature shock test. The indicator shall be subjected to a temperature shock test in accordance with method 503.1, procedure I. At the conclusion of this test, the indicator shall pass the individual tests.

4.6.18.5 Humidity test. The indicator shall be subjected to a humidity test in accordance with method 507.1, procedure I after which it shall pass the tests specified in 4.6.4 through 4.6.6. There shall be no deterioration that would affect subsequent operation.

4.6.18.6 Fungus test. The fungus test shall be conducted in accordance with method 508.1, procedure I. At the conclusion of this test, the indicator shall pass the tests specified in 4.6.4 through 4.6.6.

4.6.18.7 Salt-fog test. The indicator shall be subjected to a salt-fog test in accordance with method 509.1, procedure I after which it shall pass the tests specified in 4.6.4 through 4.6.6. There shall be no damage to the indicator that would affect its operation as a result of this test.

4.6.18.8 Dust test. The indicator shall be subjected to a dust test in accordance with method 510.1, procedure I, after which it shall pass the tests specified in 4.6.4 through 4.6.6.

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4.6.18.9 Vibration failure test. The indicator shall be subjected to a vibration failure (endurance) test in accordance with method 514.2, procedure IA, except the power spectral density (G^2/Hz) endurance test level shall be as shown in figure 5. The indicator shall be considered hard mounted in service. After completion of vibration cycling, the indicator shall be subjected to and shall meet the individual tests.

4.6.18.10 Rain test. The indicator shall be subjected to a rain test in accordance with method 506.1, procedure I. At the conclution of this test, the indicator shall pass the tests specified in 4.6.4 through 4.6.6.

4.6.19 Reliability tests

4.6.19.1 Preproduction. The six indicators shall be subjected to a reliability test in accordance with MIL-STD-781, test level F. During the ON cycle, the indicator shall be properly connected to a standard test transmitter and cycled through the complete range of the indicator at a rate of 5 units per second. The individual test shall be conducted every 100 hours of ON time.

4.6.19.2 Accept-reject criteria. Accept-reject criteria shall be in accordance with MIL-STD-781, test plan III. Upon accumulation of five failures, the indicators shall be rejected.

4.6.19.3 Longevity. Unless otherwise specified, two of the indicators specified in 4.6.19.1 shall continue to be tested for a total of 6,000 hours (3,000 hours each). Evaluation of this test shall be in accordance with MIL-STD-781, test plan XXVIII.

4.6.19.4 Acceptance. The three samples specified in 4.5.2.2 shall be subjected to an acceptance reliability test in accordance with the ON cycle conditions specified in 4.6.19.1. Each of the three samples shall be tested for 1,500 hours. At the conclusion of this test, the MTBF shall be calculated using the Chi squared distribution and shall be 1,500 hours. Each out-of-tolerance condition shall be considered a failure.

4.6.20 Maintainability demonstration. The indicator shall be subjected to a maintainability test in accordance with MIL-STD-471, appendix B, test method 1, test plan A_1 plus B_1. The indicator shall have a M_{ct} of 100 minutes less the time required to unseal and seal the instrument after repair. The $M_{max_{ct}}$ shall be no greater than 300 minutes.

4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Indicators shall be preserved and packaged in accordance with MIL-STD-794 and Method III of MIL-P-116, one each in a unit container conforming to PPP-B-636, domestic class. If not hermetically sealed, preservation method shall be selected from the table in MIL-STD-794. Electrical receptacles and connectors shall be equipped with water-vaporproof caps.

5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner that will provide protection against corrosion, deterioration, or physical damage so that serviceability is assured during shipment from supply source to the first receiving activity. This level may be the contractor's commercial practice provided the latter meets the criteria for this level as defined in MIL-STD-794.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1, shall be packed in shipping boxes conforming to PPP-B-636, weather-resistant class, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping containers shall be of uniform shape and size of minimum tare, weight, and cube.

5.2.2 Level B. Indicators preserved and packaged as specified in 5.1.1, shall be packed in domestic class exterior shipping containers, unless otherwise directed by the procuring activity. Other requirements as specified in 5.2.1 shall apply.

5.2.3 Level C. Indicators shall be packed in exterior type shipping containers in a manner that will assure safe transportation at the lowest rate to the point of delivery, and shall meet as a minimum, the requirements of carrier rules and regulations. This level may be the contractor's commercial practice provided the latter meets criteria for this level as defined in MIL-STD-794.

5.3 Marking. In addition to the standard marking requirements of MIL-STD-129, level C/C packs containing item(s)/equipment intended for immediate use at destination shall be conspicuously marked as follows on two opposite sides of the exterior container in letters as large as space permits:

FOR IMMEDIATE USE

DO NOT STORE IN THIS PACK

5.3.1 Nomenclature. Nomenclature shall be as follows:

Indicator, Angle of Attack ___#___

Applicable designation (ARU-45/A or ARU-46/A) to be entered by the contractor.

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6. NOTES

6.1 Intended use. The angle of attack indicators covered by this specification are intended for use in high-performance aircraft having an integrated panel. The display will be controlled by remote command sources that provide transmitter synchro output signals.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type indicator required (see 1.2)

c. When sampling plan B tests will not be conducted (see 4.5.2.2)

d. Levels of preservation and packaging, and packing required (see 5.1 and 5.2)

e. Special shipment marking (see 5.4).

6.3 Data. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description in the military departments' Authorized Data List (ADL). The data produced by this specification is as follows:

a. Hazard Analysis as specified in 3.4.1.4.1

b. Procedure data as specified in 3.14.1

c. Test report as specified in 4.4.2.

6.4 Asterisks are not used in this revision to identify the changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
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Preparing activity:
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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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